



CERTIFIED MATERIAL TEST REPORT
CERTIFICATE OF COMPLIANCE
CERTIFICATE # 251542
REVISION 0

Tube Forgings of America, Inc.
5200 N.W. Front Avenue
Portland, OR 97210
ISO 9001:2015 Certified
7/8/2026

Sold To	DNOW LP #MI04 1701 Gover Pkwy. Mt Pleasant, MI 48858-8140	Ship To	DNOW LP #MI04 1701 Gover Pkwy. Mt Pleasant, MI 48858-8140
Customer ID	34217		
TFA Sales Order	331068\3-1		
Purchase Order	663014514		

TFA Part ID	22232	Heat Code	342Vn ✓
Description	4 X 2 XH CONC	Grade	WPB
Quantity	1	Raw Material Heat	65153
Customer Part ID	3779	Process	WPB-CN1MIW
		Mill	TAMSA

Austenitized @:	1650°	Tempered @:	°	Stress Relieved @:		Medium:
Held For:	60 mins/min	Held For:	mins/min	Held For:	mins/min	

Heat Analysis Chemistry

C	Mn	P	S	Si	Mo	Cr	Ni	Cu	V	Cb	B	Ti	Al	N	Co	Ca	C.E.
0.17	0.98	0.009	0.004	0.19	0.02	0.11	0.07	0.22	<0.01								0.38

Tensile Test Results

Test	Name	Yield	Tensile	% Elong	R/A %	Gage Length	HBW	In-Process Test 1	In-Process Test 2	Tensile Orientation	Test Bar Type & Size
1	BASE	45900 (psi)	69200 (psi)	34.00		2"	130	125	135	LONGITUDINAL	RECTANGULAR

Charpy Impact Test Results

	Impact Energy				Lateral Expansion (Mils)			% Shear				Test Temp.	Size (mm)	Charpy Orientation
	1	2	3	Avg	1	2	3	1	2	3	Avg			

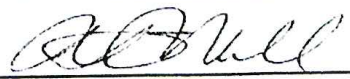
Notes/Comments

1. THESE FITTINGS WERE NORMALIZED AT 1650F AND AIR COOLED. 2. THESE FITTINGS WERE TENSILE TESTED TO VERIFY PHYSICAL PROPERTIES AFTER THE HEAT TREAT PROCEDURE. 3. THE STARTING MATERIAL IS SEAMLESS PIPE. 4. THE HARDNESS DOES NOT EXCEED 197BHN PART TO PART. 5. THIS PRODUCT IS SEAMLESS. 6. THIS PRODUCT WAS MANUFACTURED BY (MILLS IRON WORKS INC) IN ACCORDANCE WITH THE QUALITY PROGRAM REQUIREMENTS OF (TUBE FORGINGS OF AMERICA INC).

*UB = Unbroken Test

Heat: 342VN Grade: WPB

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Scot Marshall, Manager Quality Assurance

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